Batch 6 — Analysis sheet: aligning DNA reads (edit distance, Dynamic Programming)

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Table 1 — Data structures (m = n = 150 per read; 4×10^8 reads; also $10^6 \times 10^6$ genomes)

Structure	Access/update	Fill time	Space 150 × 150	Space $10^6 \times 10^6$	Edit ops?
2-D array (chosen)	O(1)	Θ(mn)	22,801 cells ≈ 22 KB	≈ 4 TB	Yes (traceback)
Hash-map memo	O(1) expected, big constant	Θ(mn)	tens of bytes per entry	> 4 TB	Yes
Two rolling rows	O(1)	Θ(mn)	≈ 300 B	≈ 8 MB	No
Recursion stack	–	$\Theta(5.83^n)$	O(m+n)	O(m+n)	By redoing work
Linear-memory version	O(1)	≤ 2mn	O(n)	≈ 8 MB	Yes

Table 2 — The five techniques

Technique	Verdict	Concrete evidence
Dynamic Programming	**Chosen**	Optimal substructure (cut off last column) + overlapping subproblems: 25 distinct subproblems; recursion makes 481 calls, (2,2) solved 13 times.
Greedy	Rejected	Counterexample ACGT vs CGT: greedy cost 4 (S S S D), true cost 1 (delete A). Greedy-choice property fails; 3 false variants.
Divide and Conquer	Rejected	Last-column split gives $T(n) = 3T(n-1) + O(1) = \Theta(3^n)$; middle split needs unknown read crossing point, sub-calls overlap. Used only inside the linear-memory version.
Backtracking	Rejected	Calls 481 / 72,958 / 12,146,179 at n = 4 / 7 / 10; $\approx 10^{113.7}$ at n = 150 (rate $3 + 2\sqrt{2} \approx 5.83$).
Branch and Bound	Rejected	Bound g + |length gap| is just g for equal lengths. Nodes 243 / 1,050 / 3,416 at n = 6 / 8 / 10 vs 49 / 81 / 121 table cells; tight bound would need the table.

Table 3 — Complexity and measurements

Item	Result
Table method	Time Θ(mn) best = worst; space Θ(mn). Fleet: $\approx 9.1 \times 10^{12}$ cell updates (≈ 2.5 core-hours at an assumed 10^9/s). Banded (cost ≤ k): O(kn).
Linear memory (Q4)	Hirschberg: space O(n), time ≤ 2mn. Measured peak memory at n = 1000: 25,513 KB vs 87 KB.
Affine gaps (Q5)	M, X, Y tables: 3 × 22,801 = 68,403 cells (3× storage, ≈ 3× time). Instance: cost 3 (other co-optimals cost 7). DP still applies.
Worst case	Input-independent. n = 1600 fill: identical 426 ms, mutated 389, random 799, A^n vs C^n 441 (same 2.57 M cells; differences from Python integer allocation).
Timing, same inputs	n = 4..11: table 0.0040 → 0.0192 ms; recursion 0.053 → 6,175 ms (≈ 320,000× slower at n = 11).
Timing, scaling	n = 100..3200 full table: 0.97, 3.65, 19.2, 173, 721, 3489 ms (≈ n^2, cost per cell rises 0.095 → 0.34 µs: cache/allocation).
Reference	BWA-MEM (Li 2013), minimap2 (Li 2018): seed first, then banded SIMD affine-gap DP near seeds.